2025

TrueRepairs

CSC 4996 - DEVELOPMENT PLAN

HISHAM ISSA, ROBERT SIMOVSKI, JIBRAN IQBAL, ISHMAEL GONZALES

Table Of Contents

1. Project Overview 3
2. Project Purpose 3
   2.1 Scope 3
   2.2 Objectives 3
3. Team Organization 4
4. Problem Resolution Policies 4
5. Project Plan 5
   5.1 Meetings 5
   5.2 Project Timeline 5
6. Configuration Management Plan 6
7. Technologies 6
   7.1 Front-End Development 6
   7.2 Back-End Development 7
   7.3 Database Technologies 7
   7.4 AI / Image Recognition 7
   7.5 Cloud / Storage Services 7
   7.6 Internal Data Source and External APIs 7

1. Project Overview

Our goal is to transform the way drivers do repairs on their vehicles by creating a multi-purpose website that integrates users' inputs, automotive databases, web-scraped parts data, and AI-driven detection features. We will deliver a website that is transparent about repair costs and accessible to everyday drivers. The integration of these features will not only provide clarity on repair costs but also demonstrate that everyone can benefit from such a tool.

---

2. Project Purpose

The project goal is to create a repair cost tool that every driver can easily understand and use. The tools can be expanded in the future, allowing for AI integration within the chassis. TrueRepairs will provide cost transparency while giving users an accurate repair estimate and difficulty. For example, users can enter their vehicle's year, make, model, or VIN along with the needed part, and the site will display the repair cost, difficulty level, and DIY price. If there is a panel on the car that is damaged, users can use the AI feature to scan the exterior and identify the damaged panel.

2.1 Scope

The scope for TrueRepairs consists of using APIs, AI detection, web scraping, and databases with a user-friendly interface to populate repair costs and DIY options. These features will be supported by vehicle databases and AI models that detect panels. Out of scope is AI detection within the chassis of the vehicle (e.g., engine bay). These features may be developed in the future but are not necessary for the current phase, which focuses on transparency in repair cost estimates.

2.2 Objectives

The project will produce a functional website for repair cost estimates using multiple integration methods such as AI models, web scraping, and APIs to deliver transparent pricing. Users will enter the vehicle's information (VIN or year, make, model), followed by issues, which will be broken down into repair data. The system will also rely on AI panel detection to identify damaged exterior parts.

---

3. Team Organization

- **Team Lead:** Hisham
- **Documentation Lead:** Hisham
- **QA/ Presentation Lead**: Robert
- **Frontend Lead:** Ishmael / Robert
- **Database / Backend Lead:** Jibran / Robert
- **AI Lead:** Hisham
- **GTA:** Kang Lu

Each team member may support multiple roles, but primary responsibility for deliverables is outlined above to ensure accountability.

---

4. Problem Resolution Policies

- **Decision Making /Voting**
  - ❖ Standard decisions will follow a 3:1 majority vote.
  - ❖ If there is a 2:2 tie, the GTA will make the final call.
- **Attendance & No-Shows**
  - ❖ Unexcused Absences:
    - ➢ 1st time: Written warning.
    - ➢ 2nd time: GTA is notified.
    - ➢ 3rd time: Professor is notified.
  - ❖ Excused Absence (with prior notice):
    - ➢ Two written notices allowed.
    - ➢ If repeated, the matter will be reported to the GTA.
- **Assignment Deadlines:**
  - ❖ Monday Night: Self Assignments due.
  - ❖ Wednesday Noon: Corrections, documentation, QA and presentation materials are due.
  - ❖ Thursday: Final Project deliverables for that week are due.
- **Meeting Schedule:**
  - ❖ Wednesdays & Fridays 12-2PM: Weekly Team Meetings on zoom.
  - ❖ Fridays, 11AM-12PM (weekly): GTA check-ins (remotely on zoom).
  - ❖ Biweekly Tuesdays starting September 16,2025: GTA Meetings (In Person in State Hall)
- **Failure to Deliver Work**
  - ❖ 1st occurrence: Written warning; teammates assist in completing work.

- 2nd occurrence: Written warning; GTA is informed; teammates help complete the task.
- 3rd occurrence: Written warning; GTA and professor formally notified that the teammate is not contributing.

---

5. Project Plan

5.1 Meetings

- **Client/Teams Meetings**
  - **Days and Times: Wednesdays from 12:00 PM – 2:00 PM; Fridays from 11:00 AM – 1:00 PM**
  - **Location:** Zoom
  - **Objectives:**
    - Conduct integration checks to ensure all components (frontend, backend, database, APIs, AI models) work cohesively.
    - Discuss relevant updates from recent lectures and apply them to project progress.
    - Plan next steps, assign weekly tasks, and track accountability.
- **GTA Meetings**
  - **Fridays, 11:00 AM – 12:00 PM (Zoom):**
    - Each team member reports updates and reviews progress with the GTA present.
    - Clarify assignment requirements and grading criteria for new deliverables.
  - **Tuesdays, 4:30 PM – 5:30 PM (In-person, before class):**
    - Provide in-person demonstrations of project functionality for immediate feedback.
    - Identify blockers hindering progress and seek guidance from the GTA.

5.2 Project Timeline

- **First Prototype (September 23rd)**
  - Functional account creation/login system with frontend, backend, and database fully connected.
  - Basic UI framework with user dashboard and page navigation.
  - Demonstrates full-stack connectivity end-to-end.
- **Software Requirements Specification (September 30th)**
- **Proof of Ethics (October 7th)**
- **Design Document (October 16th)**

- **Second Prototype (October 23rd)**
    - ❖ Expanded API integration for repair cost and parts data.
    - ❖ Frontend, backend, APIs, and database syncing verified.
    - ❖ Users can view DIY vs. shop repair cost estimates.
    - ❖ Functional repair cost outputs for entered vehicle data.
- **Third Prototype (November 11th)**
    - ❖ Nearly complete system with refined UI and smooth navigation.
    - ❖ Comprehensive repair database integration for accurate cost data.
    - ❖ AI panel detection working with real outputs, integrated into repair cost tool.
    - ❖ Enhanced account management (password reset, validation).
    - ❖ Performance tuning, robust error handling, and end-to-end testing completed.
    - ❖ System nearing final product state, requiring only polish and documentation.
- **Test Plan (November 18th)**
- **Final Project Submission (December 4th)**

---

6. Configuration Management Plan

- **Source Control:** GitHub repository for all code and documentation.
- **Branching Strategy:** Each team member will work on a personal branch.
- **Merging:** Code must be reviewed by either **two teammates** or **one teammate and the Team Lead** before merging into the main branch.
- **Version Control Practices:**
    - ❖ Commitment messages must be descriptive and linked to features or tasks.
    - ❖ Pull requests are required for merging with the main branch.
    - ❖ Conflicts will be resolved collaboratively, with Team Lead overseeing final mergers if needed.

---

7. Technologies

To successfully develop the TrueRepairs web application, the project will leverage a modern full stack architecture supported by external APIs and AI services. The following tools and technologies will be used across the development lifecycle:

7.1 Front-End Development

The application uses React.js as the front-end framework to build reusable and efficient UI components. Styling and customization are handled with CSS, while React Router supports smooth navigation between pages, and form input is managed with React Hook Form.

7.2 Back-End Development

The back end is powered by Node.js and Express.js, which handle requests and connect the front-end to the database. Secure authentication and data handling are included to protect user information.

7.3 Database Technologies

We use MongoDB Atlas, a cloud-based database, to store user accounts, vehicles, and repair reports. Mongoose is used to help organize and manage the data efficiently.

7.4 AI / Image Recognition

To identify damaged vehicle parts from uploaded photos, we use OpenAI API, which provides accurate image recognition and labeling.

7.5 Cloud / Storage Services

Vehicle images are stored and managed using Cloudinary, ensuring fast and secure retrieval when linked to reports.

7.6 Internal Data Source and External APIs

We integrate internal data source — a locally stored JSON dataset derived from RockAuto to provide vehicle parts and repair-related information. Additionally, we utilize NinjaAPI for VIN decoding and retrieving official vehicle specifications. These tools provide reliable automotive information that supports accurate reporting.